

Analyzing Economic Factors that Affect Tech Companies' Revenue Growth

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Cover Page

-Everyone has been contributing fairly to this project and is very responsive in the text chat. We have been working together throughout the past few weeks to discuss what models, data, etc. we should be using.

-I would like to praise Sasha for going above and beyond because he helped us tremendously when it comes to brainstorming the ideas and the variables we should be using, which were the most difficult part of the project. Honestly, everyone in the team has been very helpful. Hamza helps us a lot with collecting the data from various sources and put them down together in one excel file, while Alexa contributed a lot to making the slides.

Objective

We aim to conduct an analysis of the financial performance of the top 10 technology companies with highest market cap, focusing on **revenue growth** and its impact on influential factors.

Why is this topic important?

Studying the growth of the technology sector in conjunction with other economic factors provides a comprehensive understanding of its role in shaping economies, societies, and global dynamics.

Our Data

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
1	Date	CPI	Real GDP Growth	Interest Rates	Unemployment Rate	Domestic Inflation	Consumer Confidence	Subsidies for	MSFT	AAPL	NVDA	TSM	AVGO	ASML	ORCL	CRM	AMD	SAP
2	1993	2.96965	4.92553	-14.17416	-7.78643	9.26147	7.17388	22.28283	36.03	12.57					27.52			
3	1994	2.5956	4.78396	39.01296	-11.70084	13.52146	11.44439	-11.38554	23.87	15.19					33.16			
4	1995	2.80542	4.8341	38.91313	-8.33333	4.85519	-0.0271	7.20567	27.7	20.38					48.26		14.52	
5	1996	2.93667	4.77714	-9.2233	-3.27869	8.69574	1.48162	1.16225	46.05	-11.11					42.35		-20.88	
6	1997	2.33757	5.32016	3.05127	-8.62866	11.42036	10.2644	-4.38071	37.65	-27.99	230.96				34.59		20.65	-17.17
7	1998	1.54701	5.26964	-1.9536	-8.93761	8.84122	1.35637	7.63469	27.87	-16.1	643.12				25.68		7.88	40.3
8	1999	2.19314	5.89098	-7.16065	-6.2963	8.65956	1.15501	24.65898	29.39	3.25	444.31				23.56		12.41	18.22
9	2000	3.36668	6.72613	25.46948	-5.92885	8.02036	1.65367	1.17463	16.25	30.14	136.67				15.9		62.52	22.45
10	2001	2.81663	5.90663	-37.65869	19.53782	-5.08075	-17.04237	12.69368	10.19	-32.82	96.33	-28.02		-43.53	7.14		-16.2	17.27
11	2002	1.59567	4.39781	-57.12755	21.96837	-2.2866	38286	-20.71851	12.13	7.07	86.26	30.13		45.49	-11.75		-30.7	1.03
12	2003	2.298	4.53919	-32.35	3.60231	5.00608	-2.18605	20.98508	13.47	8.1	39.43	27.73		-5.48	-2.05	127.55	30.48	-5.49
13	2004	2.66757	5.31229	19.66001	-7.51043	12.54025	8.64479	-6.03544	14.44	33.38	-4.53	35.62		72.07	7.19	88.31	42.11	7.08
14	2005	3.36583	5.76531	138.17171	-8.27068	11.11087	-6.98529	31.55647	8.02	68.27	10.26	-1.18		-10.61	16.18	83.68	16.94	13.28
15	2006	3.22209	5.41692	54.48651	-9.34426	6.55829	-1.40222	-15.60151	11.29	38.65	18.19	21.61		57.72	21.87	75.68	-3.78	10.53
16	2007	2.87055	4.77069	1.10794	18083	-1.03461	-1.97576	-7.12804	15.45	27.25	29.17	2.19	9.15	16.87	25.15	60.43	4.11	9.12
17	2008	3.81495	3.9241	-61.59721	25.63177	-7.31	-25.5112	4.49981	18.19	52.54	33.53	2.04	11.26	-25.27	24.64	50.61	-0.85	12.72
18	2009	-3.2036	2.24212	-91.69909	60.05747	-22.11603	3.93464	14.82891	-3.28	14.44	-16.42	-8.88	-12.65	-44.4	3.66	43.82	-6.97	-7.72
19	2010	1.63657	2.66213	9.375	3.5009	12.22023	8.42661	-4.73872	6.93	52.02	-2.87	55.65	41.04	163.35	15.34	21.25	20.19	16.8
20	2011	3.13965	3.6994	-41.90476	-7.02515	7.71607	-6.25217	9.62303	11.94	65.96	6.52	-2.05	11.61	21.38	32.82	26.93	1.14	14.19
21	2012	2.07319	3.64816	37.70492	-9.60821	12.39801	13.64761	-3.16589	5.4	44.58	12.83	23.71	1.2	-14.68	4.21	36.77	-17.45	13.97
22	2013	1.46597	3.55229	-23.21429	-8.87513	8.26063	3.48394	2.89165	5.6	9.2	7.06	14.59	6.6	15.57	0.16	34.58	-2.27	3.66
23	2014	1.61546	3.629	-17.05426	-16.30804	8.30258	6.20726	-2.78338	11.54	6.95	-3.51	20.32	69.4	-1.77	2.95	33.47	3.91	4.44
24	2015	1.2114	2.83871	48.59813	-14.34371	6.97794	10.48044	-1.59257	27.86	13.37	6.56	59.85	-3.63	-0.13	32	-27.52	18.4	4.4
25	2016	1.26736	2.8247	198.11321	-7.58294	-30945	-1.18354	8.00307	-2.59	-7.73	7.01	14.7	94.02	5.95	-3.08	24.07	8.22	6.1
26	2017	2.13144	3.66807	153.5865	-10.59829	5.77612	5.36249	-3.09446	5.94	6.3	37.92	11.95	33.2	48.65	2.01	26.54	21.63	6.34
27	2018	2.439	4.31755	82.8619	-10.70746	7.41439	1.65346	5.7704	14.28	15.86	40.58	2.42	18.21	16.96	4.21	24.93	23.26	5.32
28	2019	1.81326	3.79095	17.83439	-5.56745	4.50043	-2.42291	15.32362	14.03	-2.04	20.61	5.46	8.39	5.58	0.31	26.02	3.95	11.51
29	2020	1.24919	3.34126	-82.58687	120.18141	-3.70076	-15.04601	806.93121	13.65	5.51	-6.81	34.07	5.71	28.36	-1.11	28.73	45.05	-0.78
30	2021	4.68219	6.61024	-78.71397	-33.8826	12.48278	-4.81349	-26.9124	17.53	33.26	52.73	20.24	14.91	24.06	3.61	24.3	68.33	1.84
31	2022	7.99169	9.2081	2,004.16667	-32.08723	12.81696	-24.01761	-73.56663	17.96	7.79	61.4	28.68	20.96	6.51	4.84	24.66	43.61	10.89
32	2023	4.12827	5.92868	198.46535	-22936	1.83423	10.80967	-20.81503	6.88	-2.8	0.22	-4.61	7.88	35.94	17.7	18.35	-3.9	1.09
33																		
34																		
35																		
36																		
37																		

Call:

```
lm(formula = GR_MSFT ~ CPI + NGDP + IR + UNRATE + GPDI + CCI +  
    GSB, data = ts_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-12.749	-5.661	-2.876	4.207	27.119

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.85775	9.17415	0.202	0.841
CPI	-0.70876	3.22260	-0.220	0.828
NGDP	3.41520	2.63628	1.295	0.208
IR	-0.28128	0.23735	-1.185	0.248
UNRATE	0.18841	0.24971	0.754	0.458
GPDI	0.80154	0.61026	1.313	0.202
CCI	0.10888	0.35011	0.311	0.759
GSB	-0.01519	0.03671	-0.414	0.683

Residual standard error: 10.67 on 23 degrees of freedom

Multiple R-squared: 0.3035, Adjusted R-squared: 0.09149

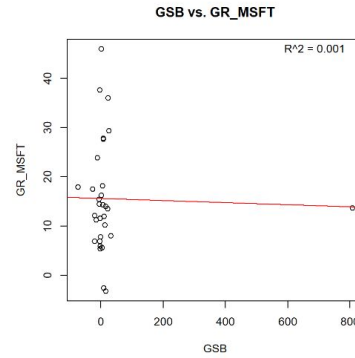
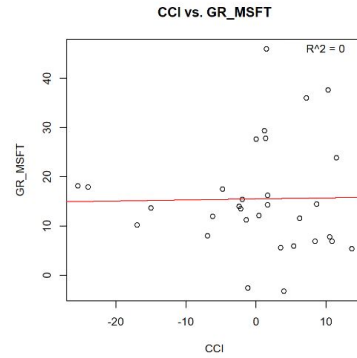
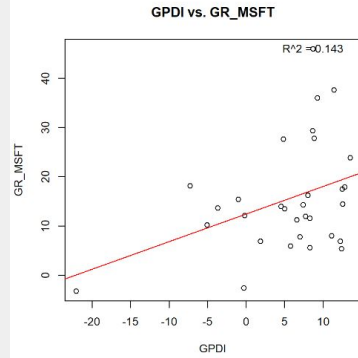
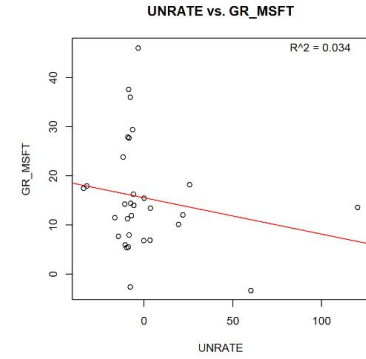
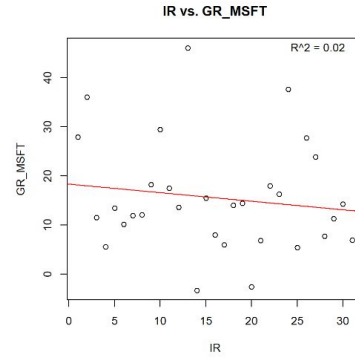
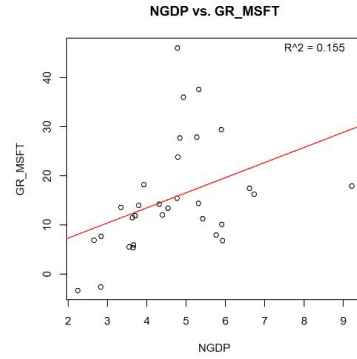
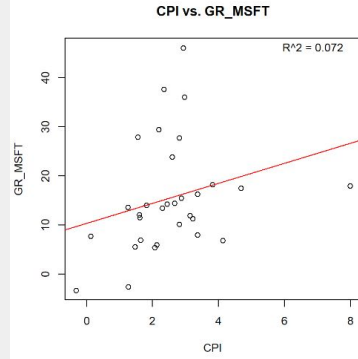
F-statistic: 1.432 on 7 and 23 DF, p-value: 0.2408

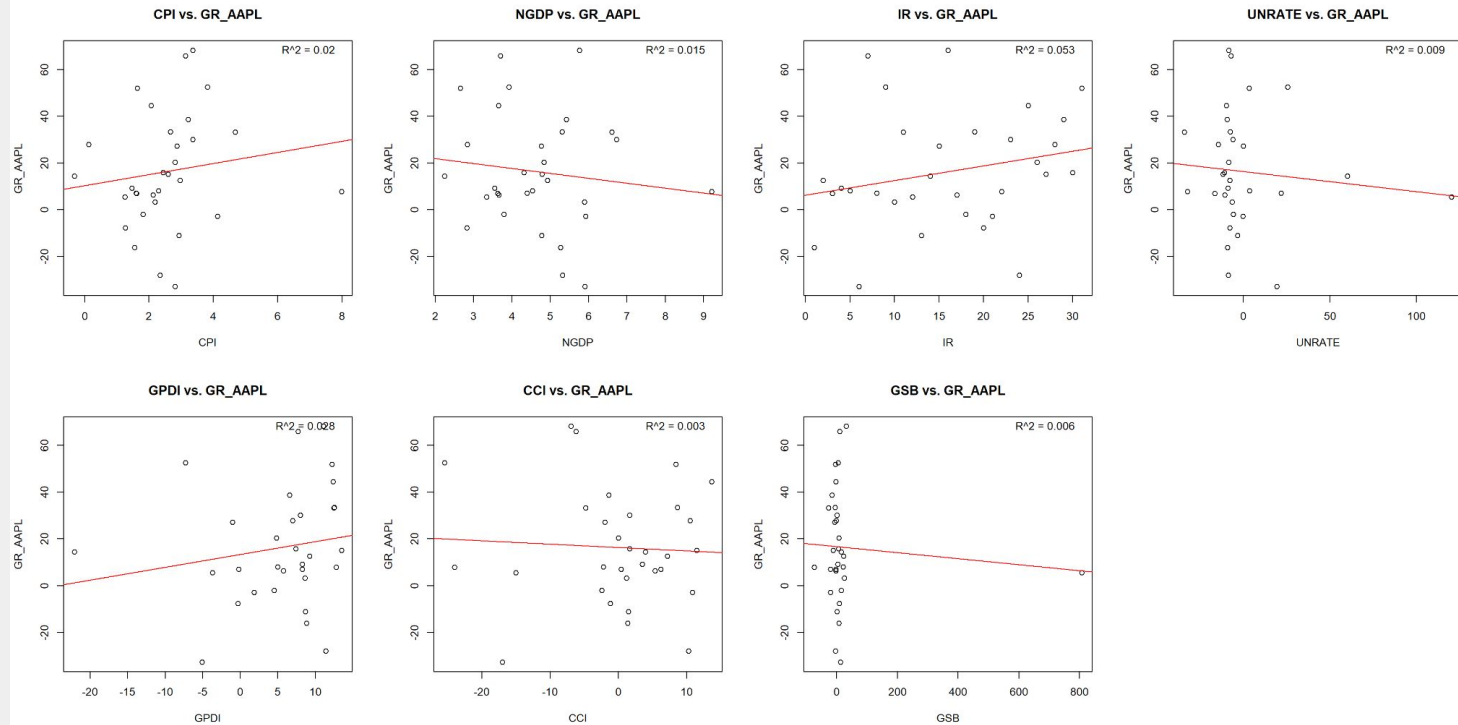
Regression
coefficients for
Microsoft

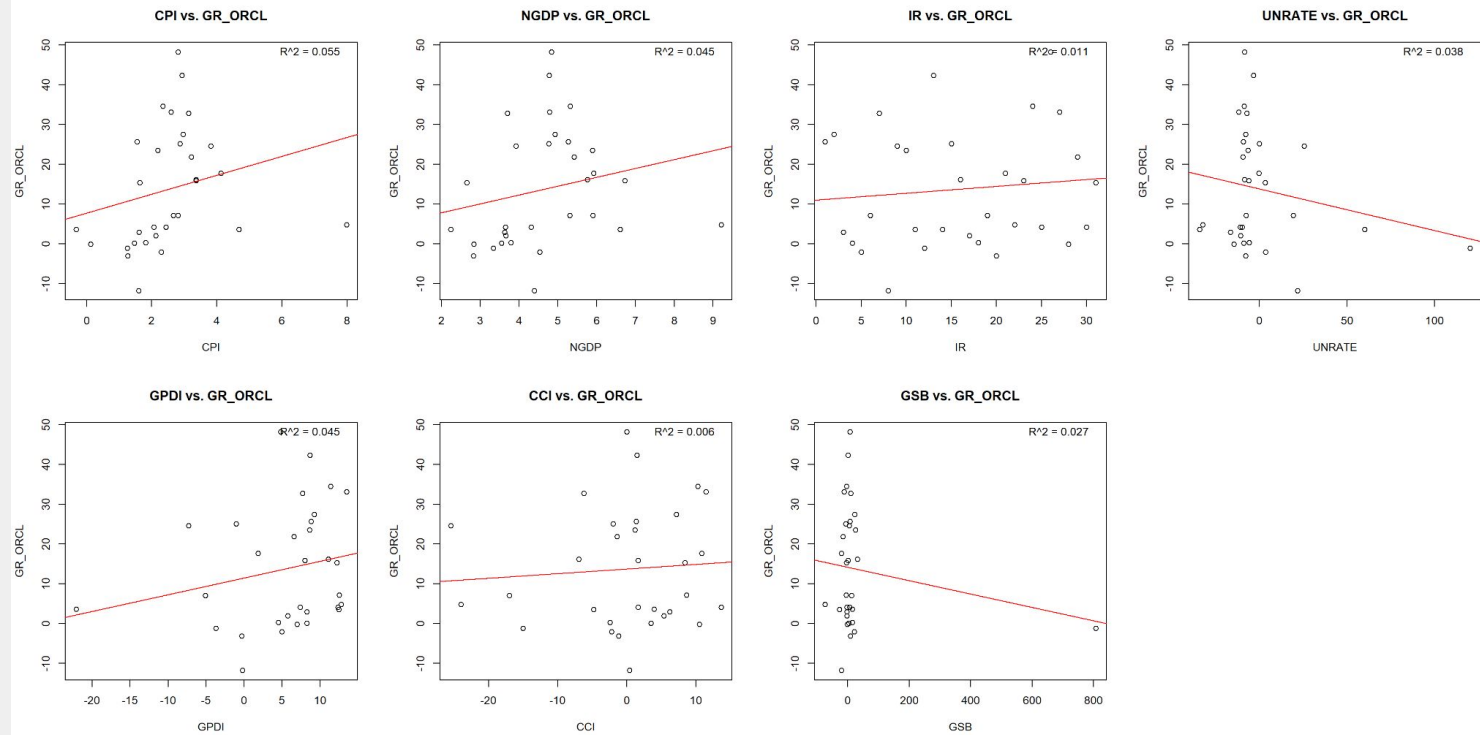
Model:

We used the **linear regression model** to find the relationship between these economic factors and revenue growth of the tech companies.

By examining historical data, regression models can help identify patterns and trends in the data, providing empirical evidence of how changes in one variable (such as GDP or private investment) relate to changes in the revenue growth rate of the tech sector.







Results and Findings

While tech companies aren't completely immune to economic downturns, these factors often make their revenue growth more resilient and less directly correlated with traditional economic indicators.

Next Steps

- Continuously monitor and update the regression model as new data becomes available to adapt to changing economic conditions and improve its predictive accuracy over time
- Drop consumer confidence index and government subsidies as they are not really relevant for this model
- Use the normal unemployment rate and interest rates instead of their percentage change